

Tabular Population Priors as Structured Retrieval for EHR Medication Set Prediction

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Abstract

Medication prediction from electronic health records is a fixed-vocabulary structured prediction problem, yet recent clinical RAG systems often treat it like open-ended case-based reasoning: retrieve similar patient admissions and ask an LLM to infer a medication set. This retrieval object is indirect—the model receives retrieved patient records while the target is a discrete set of medication-class labels. We propose a more task-aligned retrieval object: a tabular population prior—ranked medication-class candidates from a sparse condition–medication co-occurrence table estimated from training admissions. Using PACE-RAG as a fixed downstream pipeline and replacing only its patient retriever, this drop-in swap improves F1 from 0.303 to 0.338 (+11.5% relative) and Jaccard from 0.218 to 0.237 on 1,004 held-out MIMIC-IV admissions, with no additional LLM calls and no inference-time patient index. A non-LLM top- k baseline reaches F1=0.248, confirming the prior is predictive before LLM integration. Aggregate population statistics can provide a more task-aligned, efficient, and auditable retrieval source than patient-level nearest neighbors for fixed-vocabulary EHR set prediction.

1. Introduction

Electronic health records (EHRs) are a canonical structured-data domain: diagnoses, medications, procedures, laboratory events, and admissions are stored as longitudinal tables (Johnson et al., 2023). Medication recommendation from EHRs is therefore naturally framed as fixed-vocabulary set prediction: given the current admission and available patient history, a system predicts a subset of medication classes from a predefined vocabulary. Unlike open-ended clinical

text generation, the inputs are structured, the outputs are discrete labels, and performance is measured by set overlap with observed medication classes.

Medication recommendation has traditionally been studied using task-specific multi-label predictors over longitudinal patient records. Early neural methods such as LEAP model the drug set sequentially from the current patient state (Zhang et al., 2017), while later approaches incorporate external memory, graph structure, drug–drug interaction constraints, and molecular information to improve recommendation quality and safety (Shang et al., 2019; Yang et al., 2021; Wu et al., 2022; Yang et al., 2023). Recent work also explores retrieval or gating over medication history for drug recommendation (Meng et al., 2024). These models directly learn a mapping from patient histories to medication sets.

Clinical retrieval-augmented generation (RAG) systems take a different route: they retrieve evidence and supply it to a language model for grounded prediction. RAG was originally developed for knowledge-intensive NLP tasks (Lewis et al., 2020), often using dense retrieval and vector search (Karpukhin et al., 2020; Johnson et al., 2021); clinical variants retrieve guidelines, clinical knowledge, patient records, or treatment-relevant evidence for clinical question answering, medical reasoning, and personalized treatment recommendation (Zakka et al., 2024; Zhao et al., 2025; Liu et al., 2025). PACE-RAG is the most directly relevant example for our setting: it retrieves similar patient admissions using FAISS and passes those admissions through a multi-stage LLM pipeline for medication recommendation (Huh et al., 2026). This design raises a structured-data question that has received less attention: for fixed-vocabulary EHR set prediction, is the most useful retrieval object a small set of individual nearest-neighbor patients, or an aggregate summary of population-level prescribing patterns?

We study this question on admission-level medication-class prediction from MIMIC-IV (Johnson et al., 2023). Instead of indexing patient records, we retrieve from a compact *structured population prior*: a training-set condition–medication table that maps current diagnoses to ranked medication-class candidates. We use these candidates in place of FAISS-retrieved patient records in PACE-RAG, while keeping the LLM backbone, downstream stages, med-

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ication vocabulary, output parsing, and evaluation fixed. This changes the evidence given to the LLM: individual nearest-neighbor admissions are replaced by aggregate condition–medication statistics aligned with the fixed output vocabulary. On 1,004 held-out MIMIC-IV admissions, the population-prior swap improves PACE-RAG F1 from 0.303 to 0.338 and Jaccard from 0.218 to 0.237. A non-LLM top- k prior baseline reaches F1=0.248, indicating that the prior is predictive on its own, while the LLM pipeline adds further gains.

Our contributions are: (i) a sparse condition–medication population prior for EHR medication set prediction; (ii) a controlled retrieval-source swap inside PACE-RAG with the downstream pipeline held fixed; and (iii) sensitivity and stratified analyses showing when population-prior retrieval is most helpful.

2. Method

We formalize the task and the structured population prior, then describe how it replaces patient-level retrieval inside PACE-RAG.

2.1. Task formulation

For each admission i , we observe a set of current diagnoses $D_i = \{d_{i1}, \dots, d_{im}\}$, available prior-admission history H_i including recent medication history when present, and a target medication set $Y_i \subseteq \mathcal{A}$, where $\mathcal{A}$ is a fixed medication-class vocabulary. The goal is to predict $\hat{Y}_i \subseteq \mathcal{A}$ from the structured admission context. Predictions are evaluated as medication-class sets, so both missing true medication classes and over-generating irrelevant classes reduce performance.

2.2. Structured population prior

The population prior is a sparse condition–medication memory estimated only from training admissions under the patient-level split. For each training admission i , we construct a diagnosis set D_i and an observed medication-class set Y_i . For each diagnosis d and medication class a , we compute diagnosis support, pair support, and empirical conditional probability:

$$n(d) = |\{i : d \in D_i\}|, \quad n(d, a) = |\{i : d \in D_i, a \in Y_i\}|,$$

$$\hat{p}(a | d) = \frac{n(d, a)}{n(d)}. \quad (1)$$

We also compute the marginal medication frequency

$$\hat{p}(a) = \frac{|\{i : a \in Y_i\}|}{N_{\text{train}}},$$

where N_{train} is the number of training admissions, and use it to estimate base-rate lift,

$$\text{lift}(d, a) = \frac{\hat{p}(a | d)}{\hat{p}(a)}.$$

The memory stores diagnosis support, diagnosis–medication pair support, empirical conditional probability, and lift. Low-support pairs are discarded to reduce noise from rare co-occurrences, and for each diagnosis we retain the highest-ranked medication classes according to conditional probability and lift.

2.3. Inference-time aggregation

At inference time, the population prior does not retrieve patient records. For a target admission with current diagnosis set D , we aggregate diagnosis-level statistics into a medication score:

$$s(a | D) = \frac{1}{|D|} \sum_{d \in D} \hat{p}(a | d) \mathbf{1}[(d, a) \in \mathcal{I}], \quad (2)$$

where $\mathcal{I}$ is the retained condition–medication index after support filtering. To reduce the dominance of very common medication classes, we also compute a lift-normalized score over retained medication candidates:

$$\ell(a | D) = \frac{s(a | D)}{\hat{p}(a)}.$$

This is the multi-diagnosis analogue of $\text{lift}(d, a)$: it replaces the single-diagnosis conditional probability $\hat{p}(a | d)$ with the diagnosis-aggregated score $s(a | D)$ and normalizes by the marginal frequency $\hat{p}(a)$. Since candidates are drawn from retained training-set condition–medication pairs, $\hat{p}(a) > 0$ for all retrieved classes. We rank retained medication candidates by $\ell(a | D)$ and pass the top k candidates to the downstream pipeline.

The retrieved context is then a ranked list of medication-class candidates with coarse probability and lift information. It contains no nearest-neighbor admissions, patient narratives, or raw training records, so the retrieved object is directly aligned with the fixed medication vocabulary used for evaluation.

2.4. Drop-in retrieval swap inside PACE-RAG

PACE-RAG retrieves similar patient admissions using FAISS-style vector search (Johnson et al., 2021) and passes the retrieved records to a multi-stage LLM pipeline (Huh et al., 2026). We replace this retrieval call with the ranked candidates produced by the population prior. These candidates are formatted into the same context schema expected by the downstream PACE-RAG stages, while concept extraction, treatment summarization, recommendation generation, output parsing, and evaluation remain unchanged.

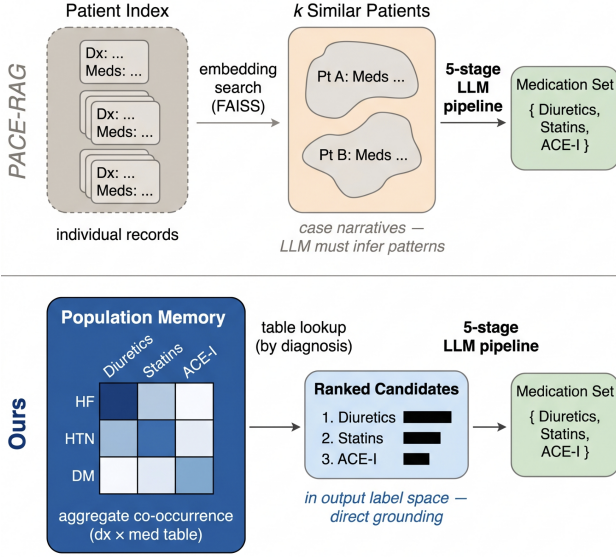


Figure 1. Retrieval contrast. Both methods index the same training records (diagnoses + medications). PACE-RAG stores individual patient records and retrieves k similar cases via FAISS embedding search. Our method aggregates these into a condition–medication co-occurrence table; a lookup by current diagnoses yields ranked medication-class candidates, which replace the FAISS-retrieved records as input to the PACE-RAG pipeline.

3. Why Population Priors for EHR RAG?

Patient-level retrieval and population-prior retrieval expose different evidence to the LLM. PACE-RAG retrieves a small set of individual admissions judged similar by an embedding index; the LLM must then infer prescribing patterns from those examples and map them to the fixed medication vocabulary. In contrast, the population prior retrieves aggregate condition–medication statistics computed over all relevant training admissions and already expressed in the medication-label space.

This distinction is useful for fixed-vocabulary EHR set prediction. Medication classes often reflect broad prescribing regularities associated with diagnoses and care contexts, while nearest-neighbor sets can be small, idiosyncratic, or sensitive to embedding quality. The population prior pools signal across the training table and is easier to audit, since each retrieved candidate can be traced to support, conditional probability, and lift. The goal is not to replace clinical reasoning with counts, but to test whether such statistics provide a better retrieval interface for the downstream LLM pipeline.

4. Experiments

Data and prediction task. We evaluate on a processed MIMIC-IV medication recommendation benchmark (Johnson et al., 2023). Each example corresponds to an inpatient

Table 1. Main results on 1,004 held-out MIMIC-IV admissions. PACE-RAG and our variant use the same GPT-4o-mini backbone; only the retrieval source differs. 95% CIs are computed via bootstrap (2,000 resamples).

Method	Retrieval source	F1 $\pm$ 95% CI	Jac.
Zero-shot (GPT-4o-mini)	none	0.223 $\pm$ 0.026	0.137
Population prior	population prior only	0.248 $\pm$ 0.027	0.148
PACE-RAG (GPT-4o-mini)	FAISS patient records	0.303 $\pm$ 0.018	0.218
PACE-RAG + prior	population prior	0.338 $\pm$ 0.017	0.237

admission with current diagnosis strings, observed medication classes, and a patient identifier used to construct chronological prior-admission history. The split is patient-level: admissions from patients used to build the population prior are excluded from the test set. The held-out test set contains 1,004 admissions. We evaluate predictions as medication-class sets using admission-averaged F1 and Jaccard similarity. This task should be interpreted as medication-class prediction from structured EHR context.

Compared systems. We choose PACE-RAG as the comparison target because it is a recent RAG-based medication recommendation system reporting state-of-the-art results on MIMIC-IV, and its public implementation exposes a patient-level FAISS retrieval module that can be replaced while keeping downstream LLM stages fixed (Huh et al., 2026). We compare four configurations. **Zero-shot GPT-4o-mini** receives only the current diagnoses and medication vocabulary, with no external retrieval. **Population prior, no LLM** directly predicts the top-ranked medication classes from the structured population prior, providing a diagnostic baseline for the signal contained in the aggregate table alone. **PACE-RAG** (Huh et al., 2026) retrieves similar patient admissions using FAISS and passes the retrieved records through its multi-stage LLM pipeline. We rerun the public PACE-RAG implementation on the same processed split used by our variant, rather than copying published numbers. **PACE-RAG + population prior** is our controlled swap: the FAISS retrieval call is replaced with a lookup into the condition–medication population prior, while all downstream PACE-RAG stages remain unchanged.

Controlled retrieval-source swap. The main comparison is PACE-RAG versus PACE-RAG + population prior under the same GPT-4o-mini backbone, temperature, medication vocabulary, output parsing, and evaluation script. This evaluates whether aggregate condition–medication statistics are a more useful retrieval object than nearest-neighbor patient records for fixed-vocabulary EHR set prediction.

5. Results

Table 1 shows that the retrieval source substantially affects PACE-RAG performance. Replacing FAISS patient-record

retrieval with the structured population prior improves F1 from 0.303 to 0.338, a +0.035 absolute gain (+11.5% relative), and improves Jaccard from 0.218 to 0.237. Since both systems share the same downstream LLM pipeline and evaluation procedure, the gain suggests that the form of retrieved evidence—aggregate medication candidates rather than neighboring patient records—is a key determinant of performance.

The remaining rows help separate three sources of performance: the base LLM, the retrieved statistics, and the interaction between retrieval and the multi-stage LLM pipeline. Zero-shot GPT-4o-mini reaches F1=0.223, indicating that diagnosis names and the medication vocabulary alone provide a weak but nontrivial signal. The population-prior-only baseline improves to F1=0.248, showing that the aggregate condition–medication table captures prescribing regularities that are not recovered by prompting the LLM without retrieval.

The full PACE-RAG + population prior system reaches F1=0.338, above both the zero-shot and prior-only baselines. This suggests that the LLM is not merely copying the highest-ranked prior candidates, but using them as structured evidence that can be filtered and reconciled with the admission context. The comparison to FAISS-based PACE-RAG further suggests that already-aggregated medication evidence is a better retrieval interface for this task than a small set of patient narratives.

Diagnostic complexity. The improvement is not driven by a single aggregate effect. As shown in Table 2, the population prior outperforms FAISS patient retrieval in every diagnostic-complexity bin. The largest absolute gain occurs for admissions with one or two current diagnoses (+0.054 F1), where nearest-neighbor retrieval may be especially brittle because the LLM must infer a prescribing pattern from only a few retrieved cases.

The gain remains positive for admissions with 10 or more diagnoses (+0.029 F1), where nearest-neighbor retrieval might be expected to be more informative. This suggests that aggregate condition-level statistics remain useful even when the diagnostic context is richer, possibly by providing a stable candidate set that the LLM can filter against the broader admission context. Inference-time latency measurements are reported in Appendix D.

Candidate-count sensitivity. We selected k on a held-out validation set of 1,000 training patients (prior rebuilt without them) and obtained $k = 30$ as the best value (val F1=0.427). Test-set F1 is essentially flat from $k = 5$ onward (range 0.338–0.340), confirming the result is not sensitive to this hyperparameter; full details are in Appendix C.

Table 2. Performance stratified by diagnostic complexity, measured by the number of current diagnoses $|D|$. The population prior improves over FAISS patient retrieval in every bin.

$ D $ bin	N	PACE-RAG F1	+Pop Prior F1
1–2	43	0.184	0.238
3–5	158	0.192	0.238
6–9	223	0.280	0.318
10+	580	0.351	0.380
All	1004	0.303	0.338

6. Discussion and Limitations

For fixed-vocabulary EHR set prediction, the retrieval object supplied to a clinical RAG pipeline matters. The population-prior variant supports the central structured-data hypothesis of the paper: aggregate condition–medication statistics can provide a more task-aligned retrieval signal than a small set of nearest-neighbor admissions.

The result also highlights a practical advantage of tabular population memory. Retrieval is a lookup into a sparse count table whose entries can be inspected through support, conditional probability, and lift, without an inference-time embedding model, vector index, or similarity search. This makes the retrieved evidence more auditable and easier to maintain while still allowing the downstream LLM pipeline to synthesize and filter candidate medication classes.

Limitations. This study should be interpreted as medication-class prediction, not prescribing advice: we do not model dose, route, timing, contraindications, drug–drug interactions, or patient-specific safety constraints. Population priors also reflect historical prescribing patterns, which may vary across institutions, time periods, and patient subgroups. External validation, recalibration, fairness analysis, and prospective clinical evaluation would be required before any deployment consideration.

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A. Population Prior Implementation Details

The population prior is a non-learned retrieval object built only from training admissions under the patient-level split. It has no trainable weights, no embedding model, and no vector index, so the controlled swap with FAISS isolates the *form* of retrieved evidence: aggregate condition–medication statistics versus patient records.

For each diagnosis string, the prior stores retained medication classes with diagnosis support, pair support, empirical conditional probability, and lift. Low-support condition–medication pairs below a fixed minimum count threshold are removed before ranking. At inference time, matching pairs for the current diagnosis set are aggregated as described in Section 2.3, ranked by the lift-normalized score, and the top k medication classes are returned. The retrieved context passed to the LLM contains only medication-class names with coarse probability or lift tiers; it does not include raw training records or nearest-neighbor patient narratives.

B. PACE-RAG Rerun and Controlled Retrieval Swap

We rerun the public PACE-RAG implementation (<https://github.com/ChaeYoungHuh/PACE-RAG>) on the same processed MIMIC-IV tables, patient-level split, medication vocabulary, and evaluation script used for the population-prior variant. We do not copy the originally reported PACE-RAG numbers, because the published implementation uses different LLM backbones (Llama-3.1-8B / Qwen3-8B) and its reported cohort statistics differ from those produced by our preprocessing. Our preprocessing of the same public codebase yielded 163,594 training patients, 342 unique ATC medication classes in the training pool, 202 in the test set, and 1,004 test admissions, compared to 196,110 patients, 406 training-pool classes, 254 test-set classes, and 1,094 test admissions in the original paper. In this paper, all LLM-based systems are run with GPT-4o-mini at temperature 0.0.

The controlled swap keeps the PACE-RAG pipeline fixed after retrieval. In the baseline, PACE-RAG retrieves similar patient admissions using FAISS and passes those retrieved records to the downstream LLM stages. In our variant, the FAISS retrieval call is replaced by a lookup into the structured population prior. The population-prior candidates are formatted into the same context schema expected by the downstream PACE-RAG stages.

Thus, all post-retrieval stages remain unchanged, and the main comparison is a retrieval-source ablation rather than a comparison between different LLM pipelines. Both our PACE-RAG baseline and PACE-RAG + population prior operate on the same processed split, backbone, medication vocabulary, and evaluation script.

C. Candidate-Count Sensitivity

We swept $k \in \{3, 5, 10, 20, 30\}$ candidates on a held-out validation set of 1,000 patients sampled by patient id from the training data. The population prior was rebuilt from the remaining training patients so that validation admissions are never in the prior used to evaluate them. Table 3 reports validation F1 used for model selection and test F1 for robustness analysis. Validation F1 increases monotonically with k , so we select $k = 30$. On the test set, F1 is essentially flat from $k = 5$ onward (range 0.338–0.340), indicating that the result is not sensitive to the exact candidate count once a small candidate set is available. The main result uses $k = 30$, the validation-selected value.

Table 3. Candidate-count sensitivity. k is selected on the validation set; test F1 is reported only to assess robustness.

k	Val F1	Test F1
3	0.319	0.330
5	0.337	0.338
10	0.367	0.340
20	0.406	0.339
30	0.427	0.338

D. Inference-Time Efficiency

Table 4. Per-admission inference latency, measured as the median over 10 sequential calls with GPT-4o-mini at temperature 0.0. Both systems run the identical five-stage LLM pipeline; the only difference is the retrieval source.

Method	Latency (median)
PACE-RAG (FAISS patient retrieval)	10.2 s
PACE-RAG + population prior	8.5 s

End-to-end latency is dominated by the same sequential LLM calls in both systems. The population prior therefore does not add LLM cost and slightly reduces retrieval overhead in this implementation. More importantly, it removes the FAISS index and embedding model from the inference path. At serving time, retrieval reduces to a lookup in a compact condition–medication table rather than maintaining a vector store and running embedding-based nearest-neighbor search.